

Traditional versus cloud computing

[00:00](#) When cloud computing was first introduced, many people resisted the idea of losing physical control over important files, cherished photos, and all sorts of other data.

[00:11](#) People were used to keeping these things close by, under their own roofs.

[00:14](#) So let's use those cherished photos as an example.

[00:18](#) Putting a photo in an album that you keep on your bookshelf does offer control, convenience, and security to a certain extent.

[00:27](#) Control can be limited by resources.

[00:29](#) You need money, materials, and time to print out a photo and purchase a frame or album.

[00:34](#) You can also only fit so many physical items in your space.

[00:38](#) As far as security goes, well, that physical memento could be damaged.

[00:42](#) Now let's think about how we can enjoy that photo if it's in the cloud.

[00:47](#) You can view and share it anytime, anywhere.

[00:49](#) And if you still want a physical copy, you can make one and feel comfort knowing that you have backup, just in case.

[00:58](#) Choosing between traditional and cloud computing is also a trade-off.

[01:02](#) Both have advantages and limitations, and both can have a place in business, depending on what the priorities are.

[01:08](#) In this video, you'll learn about traditional computing, how it works, and its defining characteristics.

[01:14](#) You'll then compare traditional and cloud computing, which will be valuable to know in the role as a cloud data analyst.

[01:21](#) So, what's traditional computing?

[01:26](#) Traditional computing is a computing model that enables data storage, access, and management through the use of physical hardware and software within a network infrastructure, typically located on-premises.

[01:37](#) Here's how that all comes together.

[01:41](#) First, hardware is set up in a dedicated space or room by IT professionals.

[01:48](#) Next, the required software —operating systems, applications, and security tools— are purchased and installed.

[01:56](#) Once the hardware and software are operational, IT personnel are responsible for maintaining and managing the entire system.

[02:04](#) This infrastructure gives an organization **sole control** and access to its data and equipment.

[02:10](#) So, with traditional computing, everything you need **is located in one location, on-premises, and can't be accessed anywhere else.**

[02:19](#) This defining characteristic offers four key advantages: control, security, compliance, and no reliance on the internet.

[02:27](#) Let's explore each of these.

[02:29](#) **First**, with traditional computing, organizations have **full control** over their hardware, software, and data.

[02:37](#) They can customize their localized network infrastructure to meet their specific needs.

[02:42](#) And **because of this control**, users usually feel more confident about the **second** advantage —**security**— if properly maintained.

[02:50](#) This is because they have sole access to their systems and sensitive information.

[02:56](#) **Third**, traditional computing might be the only viable option if a business is in an industry that requires data to be **stored on-premises.**

[03:05](#) This is an example of **compliance**, which means that a company must follow certain regulations, rules, and laws.

[03:14](#) In this case, ones **that deal with data security.**

[03:17](#) **Lastly**, traditional computing **does not rely on an internet connection** when users want to access the network or the data it contains.

[03:26](#) So, important information can be accessed even if **internet service goes down.**

[03:32](#) But just as with our photo album example, there are some downsides.

[03:36](#) **First**, with a traditional computing system, data access is limited to the device and location where the hardware and software are installed.

[03:45](#) Also, **scaling up in a traditional computing model is challenging.**

[03:50](#) **Software limitations**, the time needed to purchase and set up hardware, and the physical space required make it difficult to scale and expensive.

[04:00](#) **Besides scaling up expenses**, traditional computing involves buying hardware and software, plus ongoing maintenance of network infrastructure.

[04:09](#) **Lastly**, traditional computing can be inefficient, as each user's **software must be purchased, rather than shared**.

[04:16](#) And, again, the software is not automatically updated.

[04:21](#) These are just some of the reasons why many organizations are moving to the cloud for their computing needs.

[04:27](#) The cloud is more accessible, scalable, and offers tons of savings.

[04:31](#) It's also super-secure, efficient, and frees up staff to work on more projects.

[04:37](#) It's picture-perfect!

[04:39](#) Get it?

[04:40](#) Ha ha.